Evaluation Plan

Six-layer evaluation framework, 76-set synthetic benchmark results, real-project pilot protocol, and go/no-go gates

Six-Layer Evaluation Framework

Layer	What It Measures	Status
1. Extraction	Tags, schedule cells, dimensions, title blocks, spec paragraphs	Synthetic: benchmark · Real: 50-sheet hand check (Week 2)
2. Detection	Precision, recall & severity-weighted recall (High=3, Med=2, Low=1)	Synthetic: 76 sets · Real: scored vs past QC logs (Week 2)
3. Evidence gate	Every shipped finding cites verbatim text/coords verified in source	2,754 / 2,754 anchors verified across all 76 sets
4. Abstention	Ambiguous ('TYP.' no count) or scanned pages routed to Needs Review	0 trap violations; scanned pages downgraded / set aside
5. LLM harness	Precision, recall & 3-run consistency (aecdi llm-eval)	Harness built & self-tested; live run gated at P≥85%, C≥95%
6. Production loop	Reviewer accept rate, missed issues, review time, regression suite	Built into UI + aecdi feedback / aecdi regress

v1.2 Benchmark Splits & Results (27 Defect Kinds, 5 Traps)

Split	Sets	Defects	Det.	Shipped	FP	Prec.	Recall	Notes
Development (seed 0)	1	27	27	32	0	1.000	1.000	Used while building rules (all 27 defect kinds)
Test (seeds 1–40)	40	560	560	651	0	1.000	1.000	Error-analysis & former holdout sets
Holdout (seeds 41–60)	20	270	270	318	0	1.000	1.000	Untouched; generated & scored only in final v1.2 run
Metric (seeds 61–70)	10	143	143	166	0	1.000	1.000	Same project family drawn in mm at 1:100
Clean controls (1–5)	5	0	0	0	0	1.000	—	Zero defects — any shipped finding is a false alarm
Total (All Splits)	76	1,000	1,000	1,167	0	1.000	1.000	2,754 / 2,754 citations verified · 1.17–1.36 s/set

Holdout by Finding Type (20 Sets)

Category	Seeded	Det.	FP
Missing information	99	99	0
Cross-discipline conflict	46	46	0
Quantity mismatch	34	34	0
Spec conflict	29	29	0
Revision / redline issue	25	25	0
Dimension conflict	19	19	0
Tag conflict	10	10	0
Model vs sheet mismatch	8	8	0

Holdout Tiers

Tier	Rule Basis	Count	Correct
Confirmed	Deterministic rule on clear data	231	231
Probable	Needs interpretation (spec, scale, swing, redline)	87	87
Needs review	Abstention (bare TYP., scanned sheet)	20	—

Real-Project Pilot Protocol

Step	Method	Tool
Ground truth	3–5 past projects + QC logs & RFIs	qc_log_template.csv
Matching	Auto-match sheet/ID; human confirms	aecdi score-qclog
Extraction	50 real sheets checked by hand	Streamlit Review UI
Time saved	Timed with & without tool vs Week 1	Review log + timesheets

Pilot Go / No-Go Gates (Week 4)

Gate	Metric	Target	Status
1. Recall	Covered issues on past projects	≥ 80%	Pending pilot (synth: 100%)
2. Precision	Findings accepted by reviewers	≥ 85%	Pending pilot (synth: 100%)
3. Evidence	Shipped citations that verify	100%	Met (2,754 / 2,754)
4. False alarms	Confirmed findings on clean sets	0	Met (0 on 5 clean sets)
5. Time saved	First-pass QC time reduction	≥ 30%	Pending pilot (break-even ~21%)
6. Speed	Run time per 100-sheet real set	< 5 min	Pending (synth: ~1.3 s / 7 sheets)

Synthetic vs. real drawings

Synthetic sets prove rule consistency and evidence verification (1,000/1,000). Real-world accuracy is measured in Pilot Week 2 against Uppteam QC logs.

What earlier runs caught

v1.0 caught 5 false door tags on M-sheets & 15 missing index anchors; v1.1 caught 3 Rev-A row-shift coords & 2 loose LLM merges. All fixed in v1.2.

Deliberate traps tested

TYP. OF 4 (multiplied), bare TYP. (abstains), NIC absent from BOQ (ignored), DO NOT SCALE (probable), incorporated redlines (not flagged).

Covered vs. out-of-scope

Covers tags, schedules, specs, BOQ, dims, MEP/struct loads, revisions, redlines & Revit exports. Excludes code compliance, constructability & 3D clash.